Bayesian Probability Analysis

IMDb Movie Reviews, Sentiment via Bayes' Theorem

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1. Keyword Selection

Working as a group, we selected three keywords we believe signal positive sentiment and three that signal negative sentiment, plus one control word that appears in both classes. Every count in this report is computed directly from the dataset no values were assumed.

Positive keywords

Keyword	Justification
masterpiece	They like to use it for movies with outstanding acting.
wonderful	Reviewers like to use it when movie was exciting from start to the end.
brilliant	When script was nice and it was well directed.

Negative keywords

Keyword	Justification
boring	When movie was not interesting at all.
disappointing	When film did not meet the reviewer's expectations.
waste of time	Waste of time watching movie.

Control keyword

Keyword	Justification
funny	Appears in both praise and criticism; used as a control.

2. Choice of Conditional Probability

Our group calculates **P(Positive | keyword)** means that given that this word appears in a review, how likely is the review positive? Per the assignment, we compute this one direction only (not P(Negative | keyword)). It is the most directly interpretable framing for predicting sentiment from a keyword.

3. Dataset Statistics

Statistic	Value
Total reviews	50,000
Positive reviews	25,000 (50%)
Negative reviews	25,000 (50%)

The classes are perfectly balanced, so the prior is:

```
P(Positive) = 25,000 / 50,000 = 0.50
```

4. Keyword Frequency (reviews containing the keyword)

Keyword	In positive	In negative	Total
masterpiece	878	329	1207
wonderful	2258	522	2780
brilliant	1587	501	2088
boring	591	2460	3051
disappointing	171	615	786
waste of time	30	661	691
funny	3089	3286	6375

5. Bayesian Calculations

Formula:

```
P(Positive | keyword) = [ P(keyword | Positive) x P(Positive) ] / P(keyword)
```

For each keyword: the likelihood is P(keyword | Positive) = (count in positive) / 25,000; the marginal is P(keyword) = (total count) / 50,000; and the posterior follows from Bayes' Theorem above.

Positive keyword — "masterpiece"

```
P(keyword | Positive) = 878 / 25,000 = 0.03512
P(keyword)            = 1207 / 50,000 = 0.02414
P(Positive | "masterpiece")  = (0.03512 x 0.50) / 0.02414 = 0.727
```

Interpretation: a review containing "masterpiece" is **72.7%** likely to be positive.

Positive keyword — "wonderful"

```
P(keyword | Positive) = 2258 / 25,000 = 0.09032
P(keyword)            = 2780 / 50,000 = 0.05560
P(Positive | "wonderful")  = (0.09032 x 0.50) / 0.05560 = 0.812
```

Interpretation: a review containing "wonderful" is **81.2%** likely to be positive.

Positive keyword — "brilliant"

```
P(keyword | Positive) = 1587 / 25,000 = 0.06348
P(keyword)            = 2088 / 50,000 = 0.04176
P(Positive | "brilliant")  = (0.06348 x 0.50) / 0.04176 = 0.760
```

Interpretation: a review containing "brilliant" is **76.0%** likely to be positive.

Negative keyword — "boring"

```
P(keyword | Positive) = 591 / 25,000 = 0.02364
P(keyword)            = 3051 / 50,000 = 0.06102
P(Positive | "boring")  = (0.02364 x 0.50) / 0.06102 = 0.194
```

Interpretation: a review containing "boring" is **19.4%** likely to be positive.

Negative keyword — "disappointing"

```
P(keyword | Positive) = 171 / 25,000 = 0.00684
P(keyword)            = 786 / 50,000 = 0.01572
P(Positive | "disappointing")  = (0.00684 x 0.50) / 0.01572 = 0.218
```

Interpretation: a review containing "disappointing" is **21.8%** likely to be positive.

Negative keyword — "waste of time"

```
P(keyword | Positive) = 30 / 25,000 = 0.00120
P(keyword)            = 691 / 50,000 = 0.01382
P(Positive | "waste of time")  = (0.00120 x 0.50) / 0.01382 = 0.043
```

Interpretation: a review containing "waste of time" is **4.3%** likely to be positive.

Control keyword — "funny"

```
P(keyword | Positive) = 3089 / 25,000 = 0.12356
P(keyword)            = 6375 / 50,000 = 0.12750
P(Positive | "funny")  = (0.12356 x 0.50) / 0.12750 = 0.485
```

Interpretation: a review containing "funny" is **48.5%** likely to be positive.

6. Summary of Posteriors

Keyword	Category	P(Positive | keyword)
masterpiece	Positive	72.7%
wonderful	Positive	81.2%
brilliant	Positive	76.0%
boring	Negative	19.4%
disappointing	Negative	21.8%
waste of time	Negative	4.3%
funny	Control	48.5%

Prior baseline: P(Positive) = 50%

7. Key Findings

- **Strongest positive predictor:** "wonderful" at 81.2%.
- **Strongest negative predictor:** "waste of time" at 4.3% (i.e. 95.7% likely negative).
- All three positive keywords sit above the 50% prior; all three negative keywords sit well below it.
- The control word "funny" lands at 48.5% right at the baseline. Because it appears in both praise ("funny and clever") and criticism ("tried to be funny and failed"), it carries almost no sentiment on its own, which is exactly what a control word should show.

8. Conclusion

Bayes' Theorem turns raw keyword counts into a defensible probability of sentiment. The chosen positive and negative keywords behave as expected, and the control word confirms the method is not simply labelling every frequent word as positive. The full implementation, reading all 50,000 reviews and computing every probability in Python is provided in *Bayesian_Code.ipynb*.